



NEXTAIR - multi-disciplinary digital - enablers for NEXT-generation AIRcraft design and operations

Reporting

Project Information

NEXTAIR

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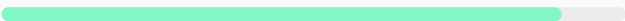
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Climate, Energy and Mobility

Total cost

€ 4 720 526,75

EU contribution

€ 4 720 525,00

Investment in EU policy priorities

Digital agenda



Clean air



Artificial
Intelligence



Climate action



Biodiversity



Coordinated by

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Periodic Reporting for period 1 - NEXTAIR (NEXTAIR - multi-disciplinary digital - enablers for NEXT-generation AIRcraft design and operations)

Summary of the context and overall objectives of the project



The aviation industry faces two challenges: maintain air traffic growth (billions of passengers each year) and reduce CO₂ emissions by 82% by 2050. Radical changes in aircraft design and maintenance are required to meet the target of climate-neutral aviation. Digital approaches are pivotal for developing new design paradigms. NEXTAIR aims to facilitate the transformation of next-generation digital aircraft by integrating advanced physical modelling with artificial intelligence. It targets the development of frugal aircraft configurations with enhanced reliability and smart maintenance. Through 8 industrial test cases, NEXTAIR validates novel design methodologies, data fusion procedures, and smart health assessment tools to address manufacturing and operational variability in innovative aircraft and engine components. These advancements will drive digitalisation in aircraft development, production, and upkeep, leading to a complete aircraft Digital Twin. This mitigates risks in designing green technologies, reducing time and cost for their development and entry-into-service

Work performed from the beginning of the project to the end of the period covered by the report and main results achieved so far



WP1: the open-source, user-friendly, customizable Multi-disciplinary Design Optimisation (MDO) engine GEMSEO has been enriched by new functionalities and plug-ins to facilitate the access of industry to the most advanced MDO techniques and to ease the setup distributed workflows. Partners have developed advanced physical models and their adjoints (e.g. for laminar flows and engine body-force). They have prepared and tested several numerical enablers to speed-up MDOs such as multi-fidelity and multi-level models, low-cost time/memory approaches to unsteady adjoint, anisotropic mesh adaptation for laminar wing design, tools for computing Pareto fronts using a reduced number of computations. Partners worked on high-dimensional Uncertain Quantification (UQ), assessing the most efficient techniques (gradient-enhanced, compressed sensing, neural network surrogates) to be employed in robust optimisation scenarios in WP5. WP2: two activities were carried out, developing new machine learning methods and assess their limitations, applicability, and performance. Partners focused on different areas such as enhancing CFD solvers with machine-learning turbulence closures, replacing disciplinary high-fidelity models with instant neural network prediction at acceptable accuracy, reconstruct real and damaged geometries from videos/images and point data, making faster their digitalization and performance analysis. WP3: partners have defined the design and optimisation problems for two high-aspect ratio wing aircraft configurations (the short-medium range DLR-F25 and a natural-laminar-flow business jet), generating the required computational models. Partners successfully set up their MDO chains and conducted preliminary optimisation studies on these test cases (newly developed capabilities from WP1 will be employed in the second part of the project). WP4: partners investigated two engine configurations aiming at drastically reducing fuel consumption of future aircraft, an Ultra-By-Pass Ratio turbofan, and an Unducted Single Fan concept. Partners carried out the setup and validation of the corresponding MDO workflows, to start

optimisation studies. Some partners also worked on the engine interaction with its environment, thanks to the introduction of a contrail microphysics model (developed in WP1) in the MDO workflow to investigate how contrail formation can be reduced by optimising the nozzle shape and the engine integration on the wing. WP5: industrial partners have defined the uncertainties that will be considered for robust optimisation problems of aircraft and engine test cases, by providing a list of the relevant uncertain parameters, their distributions, and of stochastic objective functions of interest. WP6: two activities were carried out: 1) partners draw upon the advanced methodologies developed in WP1 and WP2, to analyse and model both the manufacturing variations from new blades coming directly from industrial factories, as well as in-service ones (possibly at end-of-life); 2) topology optimisation has been used to design a heat exchanger, creating radically different flow paths than traditional serpentine designs that improve heat transfer properties, reduce pressure losses and the device mass simultaneously

Progress beyond the state of the art and expected potential impact (including the socio-economic impact and the wider societal implications of the project so far)

Results achieved so far step well beyond the state-of-art along several research directions. In the framework of advanced modelling and simulation capabilities, a new model for contrail microphysics, suitable for aircraft MDO has been developed and validated. This model provides a new discipline which was missing in the MDO framework of innovative aircraft design and engine integration although unavoidably needed (with other ones) to consider the physics of pollution in the development of new frugal concepts. New original adjoint capabilities have been developed as enablers for natural laminar flow design (thanks to continuous adjoint-to-turbulence transition models) and coupled aero-propulsive optimisations (thanks to adjoint body-force methods) that were missing in the literature while essential to prevent critical showstoppers due to inaccurate gradient prediction in the design of laminar wing shapes and innovative engine-airframe architectures. The use of Machine Learning (ML) techniques has shown a tremendous impact both for assimilation and characterisation of real data, as demonstrated on manufactured fan and turbine blades as well as at MDO level. Results obtained on representative design test cases have shown that smart DNN models can be used to replace high-fidelity physical models (e.g. turbulence, transition, heat transfer) and speed-up the resolution of MDO up to a factor 2 (also including the cost for DNN training) without damaging the accuracy of the final solution. ML models allowed to decrease by an order of magnitude (from days to hours) the time needed for digital reconstruction of in-service blades from either images or pointwise measurements, which enabled their analysis through high-fidelity CFD computations, thus, resulting in an end-to-end process for automated and fast performance characterization. The impact on aircraft engine product development cycle is potentially huge, since digitally assessing the impact of manufacturing variation (deviated blade shapes from their design-intended one) on the engine performance early on the design stage, will reduce the amount of experimental testing needs. Similar considerations hold for automatic detection and segmentation of real blade damages that have been demonstrated by using CNN, which open the way towards smart maintenance technologies. Finally, the developments carried out on the top of the MDO engine GEMSEO strongly contribute to enable and facilitate the flexible integration of high-fidelity solvers in MDO chains as well as the management of distributed

workflows, both on proprietary networks and exposed as web services. These developments ease the setup of MDO problem under uncertainty and the use of gradient-based and surrogate-based techniques for UQ through a multidisciplinary process in an efficient way, to adapt to the constraints of time-consuming industrial simulators. Therefore, the impact of these achievements could be considerable on the MDO community since these new capabilities in GEMSEO considerably reduce the MDO implementation effort (potentially from several weeks to a few days)

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